

Supplementary Clarifications and Paired-Outcome Accounting

When Valid Evidence Goes Unused in Oncology Agents

This addendum explains the submitted study and summarizes existing terminal outcomes in a post hoc descriptive analysis. We ran no new model generations. Original scores, primary contrasts, and the status of primary and exploratory analyses are unchanged. The original bibliography and Appendices A to F are included in the same PDF.

S1 Study map and endpoint definitions

The common-prompt study compares valid nonempty inventories under different supplied contexts. A separate inventory replay measures completeness on saved search paths. Case reuse and protocol differences limit comparisons across studies. They cannot be pooled as independent replications or read as successive improvements to a single intervention.

Table 1: Study relationships. Case reuse and protocol differences constrain comparisons across rows. Detailed specifications remain in archival Appendices C to E.

Study	Cases and interaction	Role in the evidence
Earlier feedback pilot	Natural and constructed proposals; original retry protocol	Identifier-only versus diagnostic feedback; preceding exploratory evidence.
Feedback × retrieval crossing	32 co-alteration and 13 sibling-disease positives; four conditions, two repetitions	Supplies the cases reused in both fixed-pool studies below.
Initial fixed exposure	Same 45 positives; ten candidates, two eligible alternatives; no search, one revision; separate negative/pass controls	Co-alteration diagnostic 11/64 versus checklist 10/64; benefit criterion fails.
Common-prompt context	Same 45 positives and candidates; five conditions, two repetitions; clarified instruction and message assembly	Primary fresh 54/64 versus diagnostic repair 43/64; other contrasts exploratory.
Length-matched live retry	Separate 24 co-alteration, 12 sibling-disease, and 12 negative cases; up to two additional searches and resubmissions	Co-alteration diagnostic 30/48 versus checklist 19/48; not an isolated effect of search access.
Inventory and replay	48 cases per arm; recall over 27 positive cases, three repetitions; replay fixes saved search paths	Measures acquired records omitted during finalization.

Valid nonempty inventory rate is the common Boolean endpoint: an answer contains at least one claim, and every claim passes the schema, visible-ID, encoded-eligibility, and copied-field checks. *Valid yield* and *valid recovery* name this endpoint in fresh and post-rejection settings, respectively. Neither implies a complete inventory. **Eligible exposure** records whether at least one eligible candidate became available. **Inventory recall** measures the fraction of the eligible reference set included in the final inventory. The 43/64 valid repair inventories and the 67.2% retrieved recall belong to different experiments and denominators.

S2 Paired outcomes and exploratory heterogeneity

We paired fresh and diagnostic-repair outputs by case and repetition index. Each pair consists of separate generations. It is not a before-and-after measurement of one trajectory, and the shared repetition index does not imply a shared model random seed. This descriptive summary supplements the original analysis, which averages repetitions within cases before variant-cluster resampling.

Table 2: Post hoc accounting of all 90 fresh/diagnostic-repair pairs from the existing common-prompt study. Invalid answers and schema failures remain distinct. Zero cells are retained.

Fresh outcome	Diagnostic repair	Co-alteration	Sibling disease
Valid inventory	Valid inventory	41	4
Valid inventory	Abstention	13	19
Abstention	Abstention	8	2
Invalid answer	Valid inventory	2	0
Schema failure	Abstention	0	1
Total pairs		64	26

The difference of eleven valid co-alteration inventories comes from thirteen pairs favoring fresh and two favoring repair. Eight pairs abstain in both conditions. Thus, not all 21 repair abstentions correspond to a successful fresh output. Fresh also produces two invalid affirmative answers, both blocked by the shared checker. Higher valid yield does not establish a uniform safety advantage. Attempted invalid answers must remain separate from released errors.

Sibling disease shows a larger exploratory difference: fresh 23/26 versus diagnostic repair 4/26, or +73.1 percentage points (reported variant-cluster interval [45.5, 93.3]). Nineteen pairs have a valid fresh inventory and a repair abstention. This stratum includes thirteen cases and eleven variant clusters. Its contrast is exploratory and unadjusted for multiple comparisons. Cases, genes, and candidate content also differ between strata; the result does not isolate a causal effect of disease rejection versus co-alteration rejection.

The primary co-alteration comparison remains +17.2 points [2.8, 33.3], based on 32 cases, 30 variant clusters, two repetitions, and 10,000 paired bootstrap draws. At the case-mean level, seven cases favor fresh, 24 tie, and one favors repair. These case-level counts and the repetition-index pairs describe different units and should not be interchanged.

S3 Generated explanations and submitted actions

The score depends on the structured action and claims, which can disagree with the submitted explanation. In an AI-assisted inspection, the authors selected the following examples from diagnostic-repair abstentions:

- In co-alteration case CQ001, repetition one, the explanation ends by recognizing two candidates as valid, while the action remains **abstain** and the claim list is empty.
- In case B032, repetition two, the explanation recognizes eligible candidates but explicitly cites concern about the preceding rejection when choosing to abstain.
- In case N0443, repetition one, the explanation recognizes encoded eligibility but raises concerns about implicit treatment-population requirements that are not encoded in the inventory predicate.

In these examples, the explanation acknowledges eligibility even though the action omits the candidates. A response can also question the clinical or source-level interpretation of a candidate that meets the encoded inventory rules. Such caution is not necessarily clinically inappropriate.

We selected these examples after inspecting outcomes and have not completed a qualitative census or estimated category frequencies. The examples do not establish an internal causal mechanism. The **reasoning** field contains generated text; it does not expose hidden reasoning. Disagreement between that text and the action does not establish that output-field order caused the disagreement.

A systematic follow-up audit should separate scope misclassification, added requirements, explicit caution after rejection, and final-explanation/action disagreement, retaining an unresolved category. There are 43 diagnostic-repair abstentions across both positive strata (21 co-alteration and 22 sibling disease). An audit could compare these abstentions with their paired fresh outputs, or cover all 180 fresh and diagnostic outputs to include successful submissions. We have not completed that audit or estimated category frequencies.

Existing audits have different targets. Archival Section C reports a 237-item enriched AI-reviewed sample from initial fixed exposure and earlier pilot/crossed explanation audits. The 211 live-retry items prepared for review have not been rated. Section E.5 reports an inventory-task qualitative sample and a post hoc census of ten failed inventories. Those studies do not constitute an audit of the common-prompt 21 co-alteration abstentions.

S4 Prompt sensitivity and identification limits

Diagnostic recovery rises from 11/64 in initial fixed exposure to 43/64 in the common-prompt study on the same co-alteration cases and candidate payloads. Three protocol changes accompany this 50.0-point descriptive difference:

1. The initial proposal appears in both the user payload and assistant call in initial fixed exposure, but only once in the later protocol.
2. The common instruction explicitly permits replacement of a rejected record.
3. It explicitly includes eligible prognostic and N/A-therapy records.

Exact initial and common instructions are disclosed in archival Sections C.4 and D.1. Because the protocol changes were bundled, the comparison cannot attribute the difference to an individual edit or quantify how much of the earlier failure came from underspecification. The recovery rate nevertheless differs substantially between the two formulations. The later fresh versus repair comparison holds the common instruction fixed and addresses a different question.

Fresh receives no erroneous proposal or rejection feedback. Repair receives both as supplied assistant/tool history. This comparison changes the proposal, feedback, message presentation, and input length as a package. Checklist and diagnostic *feedback strings* are length-matched within the specified tolerance; fresh and repair *total contexts* are not length-matched. Reset retains proposal and feedback content as user review material. The diagnostic reset and repair totals are both 43/64, but two case means improve, 29 tie, and one worsens. Equal aggregate yields do not establish equivalence, identical behavior, or validity for naturally occurring errors.

Separating the components requires new controls, such as matched-length neutral context and proposal/feedback presence varied within a fixed presentation. Restarting after removing the rejected ID

is a distinct operational policy: removing one of ten candidates also changes candidate count and input length. A matched removal of an unrelated ineligible candidate would help distinguish rejected-ID removal from generic distractor reduction. We have not run these controls, tested continuations after naturally occurring errors, or replicated the study across model versions in this update.

We froze the common-prompt analysis locally before generating its outputs, after seeing the earlier fixed-exposure results. We reused cases and payloads. Prespecification therefore concerns the subsequent execution and primary contrast, not outcome-independent selection of the research question or an independent case sample. The model is one unversioned alias. Shared publications connect all 45 positives into one component; variant-cluster intervals do not separately account for that dependence. Their coverage for generalization to new publication families is not established. A single connected component cannot supply an informative independent-component bootstrap. Additional repetitions on the same cases do not resolve these limitations.

S5 Deterministic preservation and the interpretation boundary

Let R be the deduplicated records acquired on a saved search path, P the encoded eligibility predicate, and G the full eligible reference set under that predicate. Serialization emits

$$S = \{r \in R : P(r)\} = R \cap G.$$

When identifier handling and field copying conform to the frozen contract, serialized recall therefore equals retrieved recall by construction. The empirical result is the gap between the available eligible pool and the LLM’s final inventory; the identity itself follows from the serialization rule.

On iterative scope-first paths, the LLM retains 847 of 975 available eligible record exposures and omits at least one eligible record in 29/81 positive trajectories. The 128 omitted exposures count unique identifiers within each trajectory. An identifier can recur across trajectories, so this count does not represent 128 distinct corpus records. Case-macro retrieved and final inventory recall are 67.216% and 60.302%, respectively. Their paired difference is 6.914 points, whereas 847/975 is pooled retention and uses a different denominator. Original generation has a 4,096-token response cap with no observed truncations; serialization is uncapped. The replay fixes saved retrieval and changes finalization under these output constraints.

When inventory membership is fully computable, deterministic selection and copying can preserve acquired records without an LLM. This procedure provides a direct baseline for finalization. Search formulation and source-level interpretation are possible LLM roles, but this replay does not establish their necessity, superiority, or clinical benefit. Full-corpus enumeration reaches 100% encoded recall by construction, without establishing publication-level validity.

The checker and reference labels use the same frozen curation. Agreement therefore establishes conformance to the encoded task, but independent publication support still requires source review. W04 illustrates this boundary through observations tied to different genotypes and experimental models. It is one example within an exploratory twelve-family review, not an estimate of corpus error or validation of the repair effect. Two medical-student readers used substantive AI assistance and nonblinded joint adjudication; the separate thirty-pair independent review was discontinued without completed judgments. No workflow study measured review time or decision quality. Preservation and source interpretation remain separate evaluation targets.

S6 Finding material cited in the paper

This guide refers to the 20-page submitted paper. Its condensed appendices and this supplement’s expanded appendices use different subsection and table numbers. All page numbers below are PDF viewer pages in the combined *Extended Methods and Results*. On PDF pages 2 to 32, subtract eight from the printed page number. Tables in addendum sections S1 and S2 have their own numbering.

Sections and tables

Reference in the paper	Expanded supplement	PDF pages
Table 1: protocol roles and budgets	S1 overview; C, Table 14; E.1	33, 13, 25
B, Table 3: samples and limits	B.1, B.5; study settings in C to E	7, 10 to 12
Introduction; A, Table 2: typed scope audit	A.4, Table 4	4 to 5
Section 2; B: shared methods	B.1 to B.5	7 to 12
Section 3; C.1, Table 4: fixed exposure	C.1, Table 15; outcomes in Table 17	13 to 15
Section 4.3; C.2, Table 4: live retry	C.2, Tables 16 and 17	14 to 15
C.3: initial prompts and messages	C.4	19 to 20
Section 4; D.1, Table 5: common prompt and conditions	D.1, Table 22	20 to 21
D.2: design, scoring and statistics	D.2	21 to 22
D.3, Tables 6 and 7: outcomes and contrasts	D.3, Tables 23 and 24	22 to 23
D: detailed resource accounting	D.4, Table 26	24
Section 5.1; E.1: inventory task	E.1 to E.2, Tables 27 to 29	25 to 26
E.2, Table 8: saved-pool serialization	E.4, Tables 31 and 32; replay checks	28 to 29
E.3: controls and omissions	E.3 to E.5	27 to 29
F, Table 9: joint judgments	F.3, Table 35, columns J	31
F, Table 10: W04 observations	F.3, W04 discussion	31
F: source review and its limits	F.1 to F.4, Tables 33 to 36	29 to 32

The initial system instruction, checklist core, and feedback tail appear in C.4 (PDF pages 19 to 20). The exact common system instruction and final command appear in D.1 (PDF page 21). These passages let readers compare the disclosed instructions even though the original prompt JSON files are not included. Related work is in B.6 (PDF page 12), and the bibliography is on PDF page 2.

What the public data reproduce

`data/paired_outcomes.csv` contains 90 fresh/diagnostic-repair pairs (180 outputs): 64 co-alteration pairs and 26 sibling-disease pairs. `source/analyze.py` recomputes the S2 counts. The CSV covers two of the five common-prompt conditions, whose full study comprised 450 outputs. It omits variant-cluster assignments, the other conditions, candidate records, and full responses. It therefore does not support independent response scoring, reconstruction of all five conditions, or reproduction of the reported variant-cluster bootstrap intervals. Original scoring and statistical methods are described in D.2 to D.3.

Files and documents cited in the paper

The paths in the paper and original Appendix B.5 refer to the research workspace used for the experiments. Original Table 13 is a map of that workspace, not this repository. The original experiment directories, code, and execution records are not included here. The passages below explain the methods. Running the original experiments would require the original code and inputs.

Cited file or document	Description available in this PDF
<code>versions/v14-full-paper-revision/README.md</code> ; <code>src/common.py</code> ; <code>src/agents.py</code> <code>analysis_inputs/prompt_material_exact.json</code>	B.1 to B.5, pages 7 to 12: corpus, eligibility, scoring and execution. Original README and code are not included. C.4, pages 19 to 20: disclosed initial instruction and feedback text. Original JSON is not included.
V40 <code>src/study.py</code> ; V38 E1 <code>src/runner.py</code> and <code>src/feedback.py</code>	C.1 to C.2 and C.4, pages 13 to 15 and 19 to 20: repair protocols and message assembly. Original code is not included.
<code>analysis_inputs/v41_public_prompt_contract.json</code>	D.1, pages 20 to 21: exact common system instruction and final command. The full JSON, schema and request hashes are not included as files.
<code>versions/v41-repair-history-control/</code>	D.1 to D.4, pages 20 to 24: study design and aggregate results. Original experiment directory is not included.
Other directories, execution plans, timing files and per-run scores listed in B.5	B.5, pages 10 to 12, and the study appendices. The original files and complete per-run results are not included.
Saved-pool replay code and inputs listed in B.5	E.4, pages 28 to 29: procedure and reported verification. Executable replay and original inputs are not included.
Companion technical report	A, C and F describe related studies and review scope. The separate report and its additional results are not included.
Signed reader submissions and source-bearing review materials	F.1 to F.4, pages 29 to 32: methods and reported judgments. Underlying restricted materials are not included.

Building the public PDFs

`source/build.py` compiles the cover and addendum, combines them with the original bibliography and appendices, and verifies that the original pages have identical text and rendering. It clears identifying PDF metadata and records checks and artifact hashes in `data/verification.json`. The original supplement is retained in `archive/`. The build requires Python 3, PyMuPDF, and `pdflatex`; it does not require the unavailable experiment files, licensed candidate text, full transcripts, or private reader forms. The README lists the commands and links to all public data and source files.